

MARIUPOL', UkraineWMO: **347120**Lat: **47.0425N**Lon: **37.4844E**Elev: **230**StdP: **14.57**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	8.8	-2.9	4.8	5.7	2.2	6.2	10.8	31.9	20.2	29.8	22.1	12.5	50	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	89.9	70.5	87.3	70.1	84.6	69.5	75.0	84.0	73.2	82.5	71.6	81.1	9.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	119.7	80.3	70.0	111.4	78.8	68.2	104.6	77.4	38.6	84.2	37.0	82.5	35.5	81.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
28.7	24.9	21.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-1.0	93.7	8.2	4.1	-6.9	96.6	-11.6	99.0	-16.2	101.3	-22.1	104.3		
			-1.7	78.3	7.9	2.1	-7.4	79.8	-12.1	81.0	-16.5	82.2	-22.2	83.7		
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	27.1	28.7	37.3	49.9	61.6	69.8	74.4	73.5	62.9	50.6	39.1	30.7
	DBStd	18.48	9.28	8.73	6.39	6.33	6.89	6.03	5.15	6.13	6.82	7.77	7.75	8.63
	HDD50	2806	710	596	395	77	4	0	0	0	3	89	333	599
	HDD65	6090	1175	1016	859	455	149	23	1	5	117	448	778	1064
	CDD50	3017	0	0	0	74	364	593	756	729	389	107	5	0
	CDD65	828	0	0	0	1	45	166	292	269	53	2	0	0
	CDH74	6915	0	0	0	9	204	1238	2716	2532	214	2	0	0
	CDH80	2037	0	0	0	1	21	291	851	846	27	0	0	0
Wind	WSAvg	10.4	11.4	11.6	11.7	10.9	9.5	9.2	8.9	9.4	10.0	10.0	10.9	11.3
Precipitation	PrecAvg	20.7	2.0	1.5	1.6	1.6	1.8	2.2	1.7	1.3	1.8	1.5	1.8	2.0
	PrecMax	33.5	7.6	5.2	3.3	5.3	5.0	5.2	6.5	4.3	7.3	3.3	3.9	4.6
	PrecMin	14.9	0.5	0.4	0.2	0.1	0.0	0.3	0.1	0.0	0.2	0.4	0.2	0.4
	PrecStd	4.7	1.4	1.1	0.8	1.1	1.3	1.3	1.4	1.3	1.5	0.7	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	46.5	55.3	72.8	82.6	89.5	93.3	96.1	82.8	72.0	58.4	49.0
		MCWB	42.1	43.0	47.3	59.2	66.4	70.2	70.1	68.8	66.1	60.2	54.8	47.4
	2%	DB	41.3	42.9	51.4	65.9	78.8	86.0	89.8	91.0	78.9	67.9	54.6	45.1
		MCWB	40.0	40.4	44.5	54.5	65.3	69.4	71.3	70.5	65.7	59.4	52.1	43.7
	5%	DB	39.2	40.3	48.4	62.5	76.0	83.4	87.3	87.6	76.2	64.7	51.9	43.0
		MCWB	38.2	38.4	43.3	53.7	63.8	68.2	70.9	69.5	64.2	58.6	49.7	42.0
	10%	DB	37.1	38.1	46.0	59.6	73.3	80.7	84.6	84.5	73.7	62.2	49.6	41.1
		MCWB	36.3	36.8	42.0	52.2	62.8	67.6	70.0	69.4	62.6	57.5	47.8	40.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	44.2	48.8	59.9	70.4	75.8	77.3	77.8	72.2	64.7	55.7	47.8
		MCDB	43.7	45.7	53.8	69.3	78.1	82.9	86.2	86.3	78.4	67.7	57.2	48.5
	2%	WB	40.4	41.0	45.9	56.7	67.8	73.0	75.2	74.6	68.9	61.7	52.6	44.1
		MCDB	41.1	42.2	49.3	63.0	74.9	80.8	85.0	83.8	75.6	65.5	54.1	45.0
	5%	WB	38.3	38.7	44.0	54.9	65.9	71.2	73.5	72.7	66.9	59.8	50.2	42.1
		MCDB	39.0	39.8	46.9	60.4	73.3	79.4	83.2	82.7	72.8	63.4	51.5	42.8
	10%	WB	36.3	37.0	42.5	53.3	63.9	69.5	71.8	71.0	64.9	57.8	48.0	40.3
		MCDB	36.9	37.9	45.1	58.1	71.2	77.8	81.6	81.1	71.1	61.5	49.1	41.0
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.9	10.0	12.5	13.8	15.0	15.8	16.0	14.4	11.8	8.9	7.9
		MCDBR	8.0	10.4	13.8	17.1	16.8	18.2	17.8	18.9	16.3	14.0	11.7	8.8
		MCWBR	7.1	8.4	9.2	10.4	10.2	11.1	10.5	11.3	10.7	9.6	9.6	8.0
	5% WB	MCDBR	7.4	9.3	12.0	14.1	14.4	15.4	15.5	16.0	13.7	12.7	10.6	8.4
	MCWBR	7.0	8.2	8.8	9.3	9.7	10.5	10.6	11.1	10.6	9.6	9.3	7.9	
Clear-Sky Solar Irradiance	taub	0.345	0.376	0.407	0.455	0.418	0.420	0.445	0.444	0.414	0.384	0.356	0.346	
	taud	2.277	2.200	2.174	2.084	2.298	2.325	2.258	2.246	2.297	2.335	2.332	2.283	
	Ebn at Noon	227	243	255	253	269	269	259	254	251	239	221	210	
	Edh at Noon	25	33	40	48	40	39	42	40	35	29	24	23	
All-Sky Solar Radiation	RadAvg	308	598	966	1497	1946	2056	2046	1836	1313	782	381	245	
	RadStd	51	103	117	141	159	137	107	121	102	103	56	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.12	N/A	N/A	N/A	N/A	N/A	-270	-270	+166	+162
Regional (5 neighbors)		+1.17	N/A	N/A	N/A	N/A	-0.90	N/A	-274	+229	+169